

Unit 4

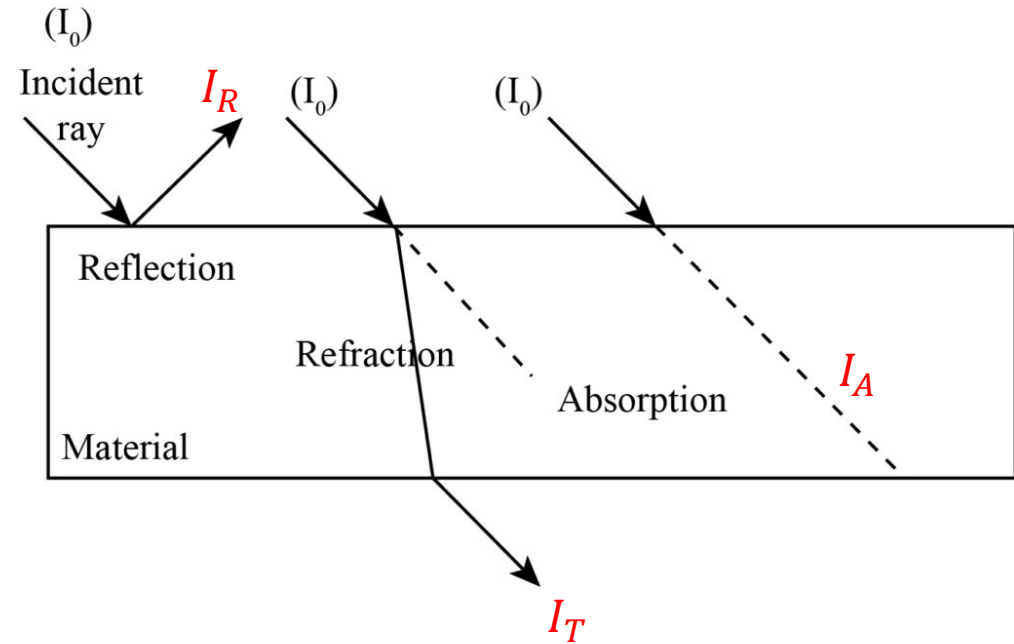
EC 208

Optical Properties

- Incoming photon give their total energy to the material → **Absorption**
- Incoming photon give their total energy to the material but photons of identical energy is emitted → **Reflection**
- Incoming photon give their fraction of total energy to the material → **Transmission**
 - Change in velocity → **refraction***
 - Change in velocity and direction → **transmission***

$$I_0 = I_A + I_R + I_T$$

$$1 = A + R + T$$



$$A = \frac{I_A}{I_0} = \text{Absorptivity}$$

$$R = \frac{I_R}{I_0} = \text{Reflectivity}$$

$$T = \frac{I_T}{I_0} = \text{Transmittivity}$$

Optical Properties

Q. A metal surface has a absorptivity of 30% and reflectivity of 50% for light at a wavelength of 500 nm. If a beam of light with an intensity of 100 W/m^2 strikes the surface, calculate the transmittivity. Also calculate the intensity of the absorbed, reflected and transmitted light.

Given,

$$A = 0.3$$

$$R = 0.5$$

$$I_0 = 100 \text{ W/m}^2$$

$$T = 1 - A - R = 1 - 0.3 - 0.5 = 0.2 = 20\%$$

$$I_A = AI_0 = 0.3 * 100 = 30 \text{ W/m}^2$$

$$I_R = RI_0 = 0.5 * 100 = 50 \text{ W/m}^2$$

$$I_T = TI_0 = 0.2 * 100 = 20 \text{ W/m}^2$$

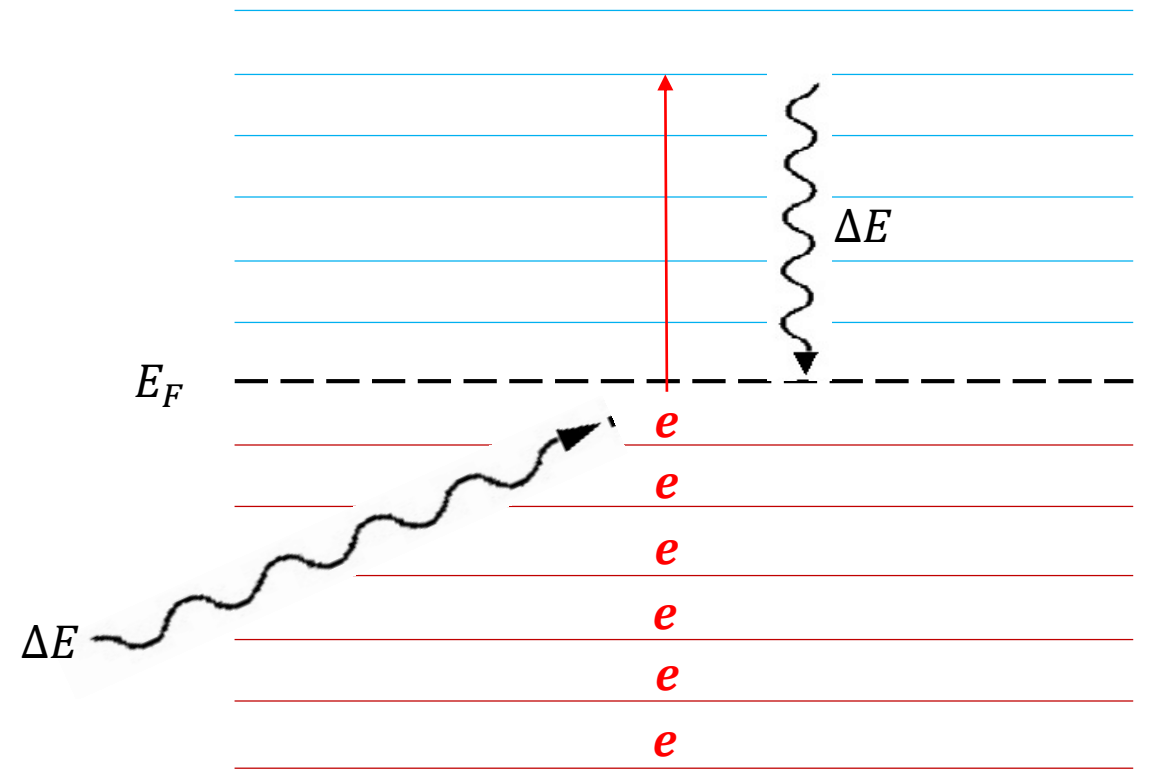
$$A = \frac{I_A}{I_0} = \textit{Absorptivity}$$

$$R = \frac{I_R}{I_0} = \textit{Reflectivity}$$

$$T = \frac{I_T}{I_0} = \textit{Transmittivity}$$

Optical Properties of Metal

- Metals do not have any band gap → it has continuously available empty states which permits electron transmission.
- Almost all frequencies of visible light are absorbed by metals.
- Metals are mostly opaque to visible light provided the thickness is in nm range or above.
- The absorbed light is mostly re-emitted from the metallic surface in the form of visible light of almost same wavelength.
- The reflectivity of metal is between 0.90 to 0.95

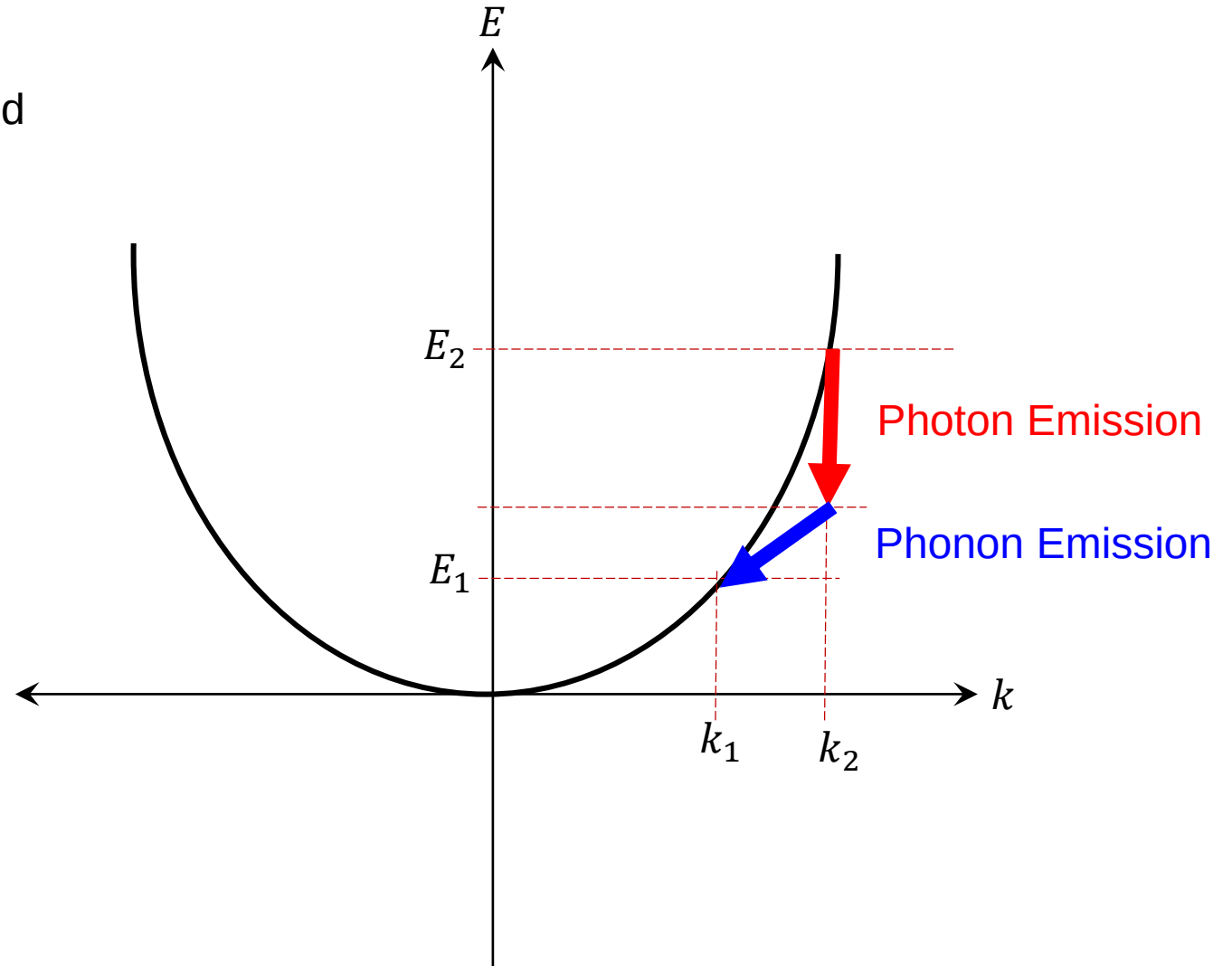


Optical Properties

- During absorption and emission, both E and k changes.
- E changes $\rightarrow$ light
- k changes $\rightarrow$ heat
- Phonon emission is necessary for change in k

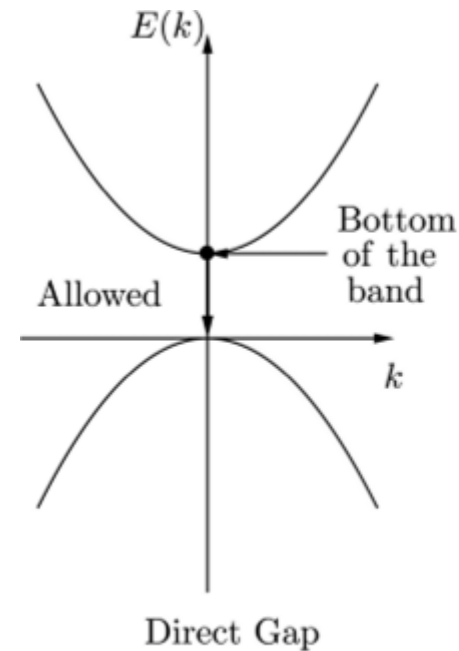
photons have **high energy** but **low momentum**

phonons have **low energy** but **high momentum**

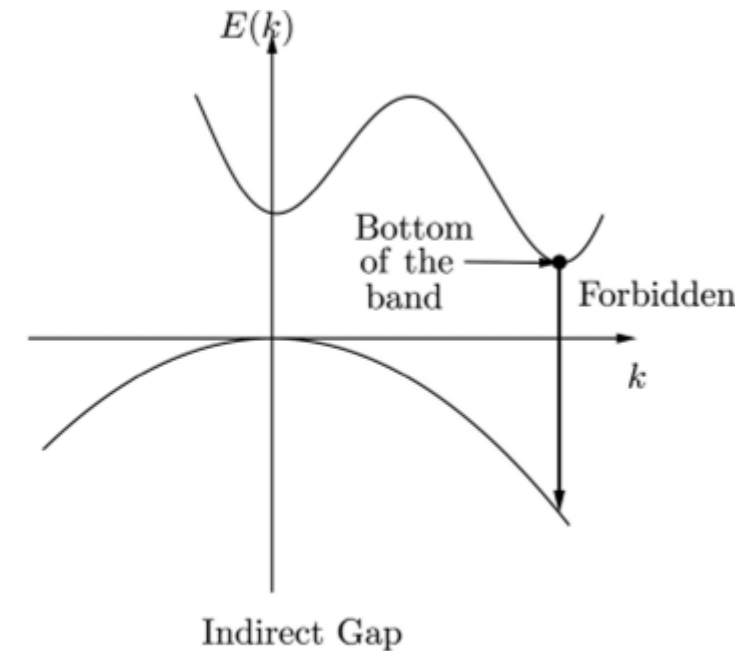


Direct vs Indirect Band Gap Semiconductors

- Direct and indirect bandgap semiconductors are distinguished by how electrons transition between the valence and conduction bands, impacting their efficiency in optical applications.
- Direct bandgap semiconductors allow for a direct transition, where the electron's energy and momentum are conserved during the process, leading to higher optical efficiency.
- Indirect bandgap semiconductors require the involvement of a phonon (a quantized vibration of the crystal lattice) to conserve momentum, resulting in lower optical efficiency.



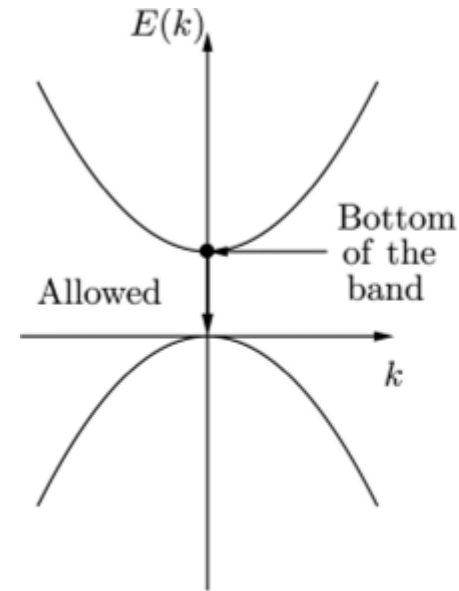
(a)



(b)

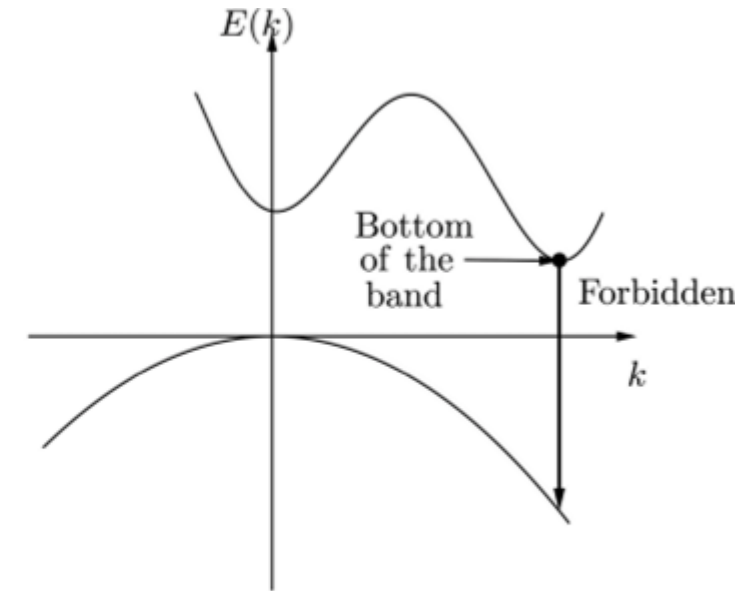
Direct vs Indirect Band Gap Semiconductors

- In direct bandgap semiconductors, the top of the valence band and the bottom of the conduction band occur at the same value of momentum (k – *value*), allowing for a direct transition without needing a phonon
- In indirect bandgap semiconductors, these points occur at different values of momentum (k – *value*), requiring a phonon to conserve momentum during the transition.



Direct Gap

(a)

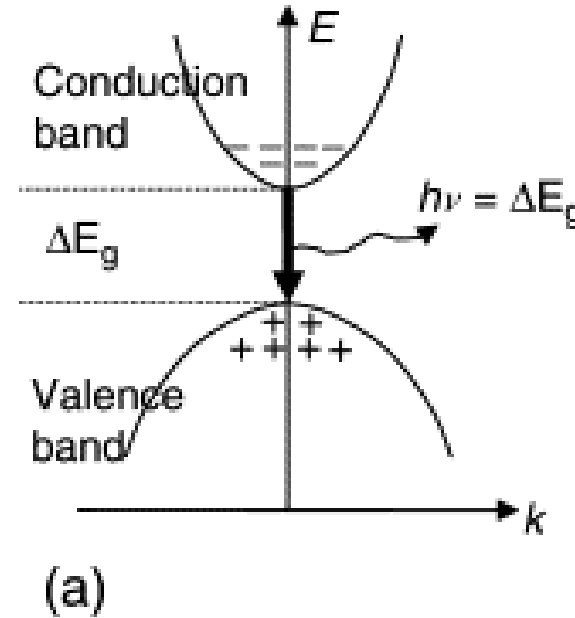


Indirect Gap

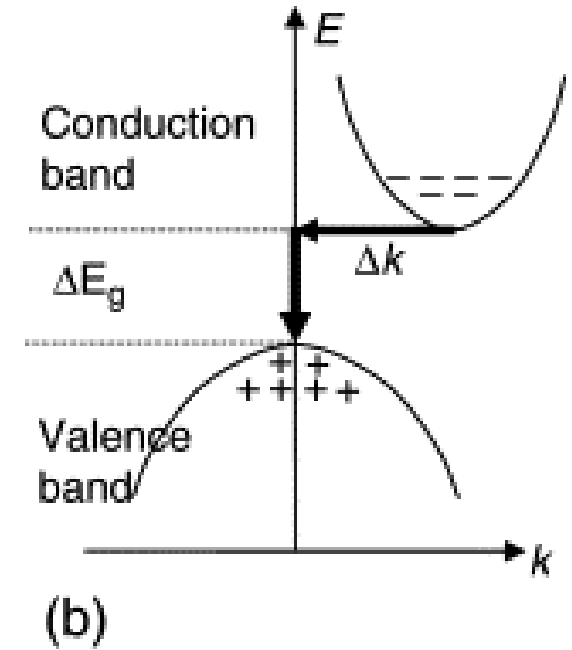
(b)

Direct vs Indirect Band Gap Semiconductors

- Direct bandgap semiconductors have higher radiative recombination efficiency (emission of light) because the electron can directly fall into the hole without needing phonon assistance.
- Indirect bandgap semiconductors have lower radiative recombination efficiency because the process is phonon-assisted, which is less probable.
- Direct bandgap semiconductors like Gallium Arsenide (GaAs) and InP are favored for optical applications like LEDs and lasers due to their high recombination efficiency.
- Indirect bandgap semiconductors like Silicon (Si) and Germanium (Ge) are less efficient for optical devices.



Photon $\rightarrow \Delta E$

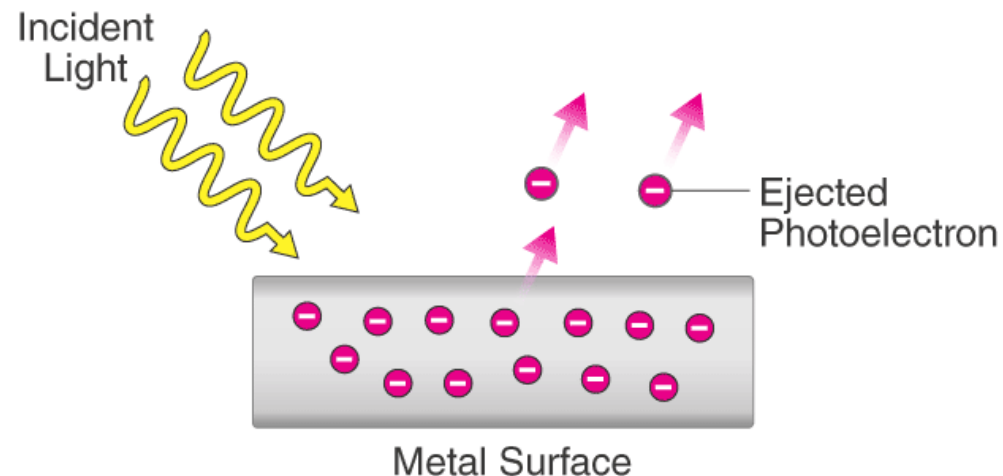


Phonon $\rightarrow \Delta k$

Photon $\rightarrow \Delta E$

The photo-electric effect

- The photoelectric effect is a phenomenon in which electrons are ejected from the surface of a metal when light is incident on it.
- These ejected electrons are called photoelectrons. The emission of photoelectrons and the kinetic energy of the ejected photoelectrons is dependent on the frequency of the light that is incident on the metal's surface.
- The process through which photoelectrons are ejected from the surface of the metal due to the action of light is commonly referred to as photoemission.
- The photoelectric effect occurs because the electrons at the surface of the metal tend to absorb energy from the incident light and use it to overcome the attractive forces that bind them to the metallic nuclei.



The photo-electric effect

The photoelectric effect cannot be explained by considering light as a wave. However, this phenomenon can be explained by the particle nature of light, in which light can be visualized as a stream of particles of electromagnetic energy. These 'particles' of light are called photons.

The energy held by a photon is related to the frequency of the light by Planck's equation:

$$E = h\nu$$
$$\nu = c / \lambda$$

Where,

E the energy of the photon

h is Planck's constant

ν the frequency of the light

c is the speed of light (in a vacuum)

λ is the wavelength of the light

Thus, it can be understood that different frequencies of light carry photons of varying energies.

For example, the frequency of blue light is greater than that of red light (the wavelength of blue light is much shorter

than the wavelength of red light). Therefore, the energy held by a photon of blue light will be greater than the energy held by a photon of red light.

The photo-electric effect

Relationship between the Frequency of the Incident Photon and the Kinetic Energy of the Emitted Photoelectron

$$E_{\text{photon}} = \Phi + E_{\text{electron}}$$

$$h\nu = h\nu_{th} + \frac{1}{2}m_e v^2$$

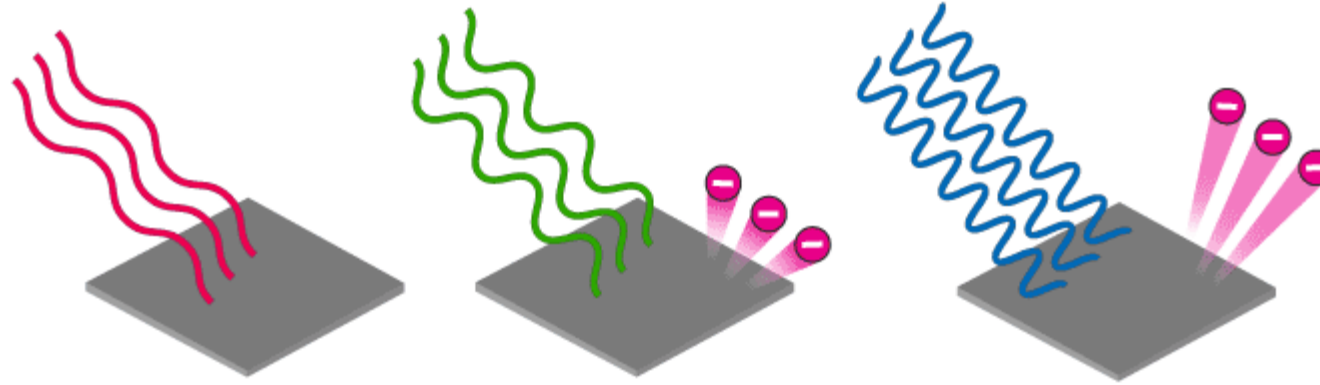
Where, E_{photon} the energy of the incident photon, which is equal to $h\nu$

Φ the threshold energy of the metal surface, which is equal to $h\nu_{th}$

E_{electron} the kinetic energy of the photoelectron, which is equal to $\frac{1}{2}m_e v^2$ (m_e = mass of electron = $9.1 * 10^{-31} \text{ kg}$)

If the energy of the photon is less than the threshold energy, there will be no emission of photoelectrons (since the attractive forces between the nuclei and the electrons cannot be overcome). Thus, the photoelectric effect will not occur if $\nu < \nu_{th}$. If the frequency of the photon is exactly equal to the threshold frequency ($\nu = \nu_{th}$), there will be an emission of photoelectrons, but their kinetic energy will be equal to zero.

The photo-electric effect



- The photoelectric effect does not occur when the red light strikes the metallic surface because the frequency of red light is lower than the threshold frequency of the metal.
- The photoelectric effect occurs when green light strikes the metallic surface, and photoelectrons are emitted.
- The photoelectric effect also occurs when blue light strikes the metallic surface. However, the kinetic energies of the emitted photoelectrons are much higher for blue light than for green light. This is because blue light has a greater frequency than green light.

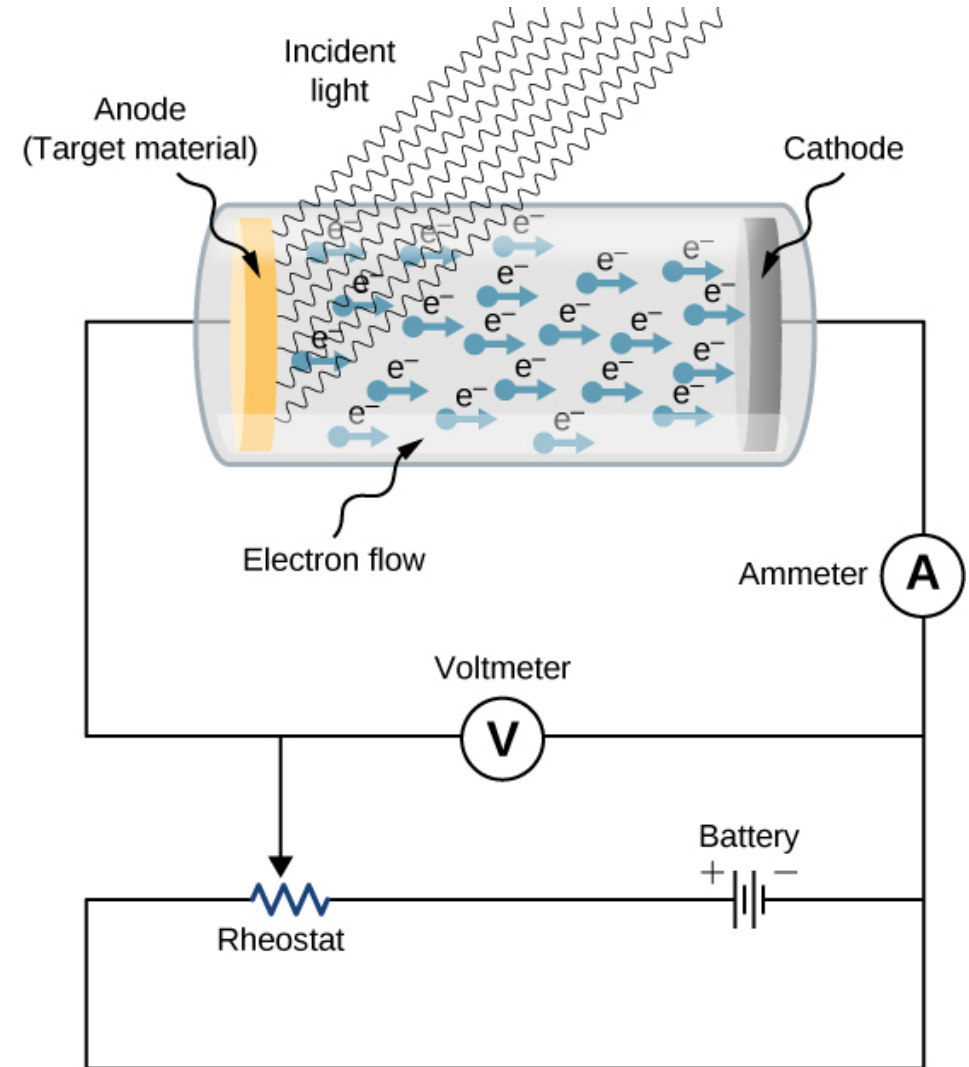
Characteristics Of Photoelectric Effect

1. The threshold frequency varies with material, it is different for different materials.
2. The photoelectric current is directly proportional to the light intensity.
3. The kinetic energy of the photoelectrons is directly proportional to the light frequency.
4. The stopping potential is directly proportional to the frequency and the process is instantaneous.

The photo-electric effect

Experimental Set-up

1. Quartz vacuum tube.
2. Two metallic electrode separated by small distance.
3. Anode is wider to cover maximum possible area in front of cathode.
4. Variable voltage source.
5. Ammeter
6. Light source whose intensity and frequency can be controlled

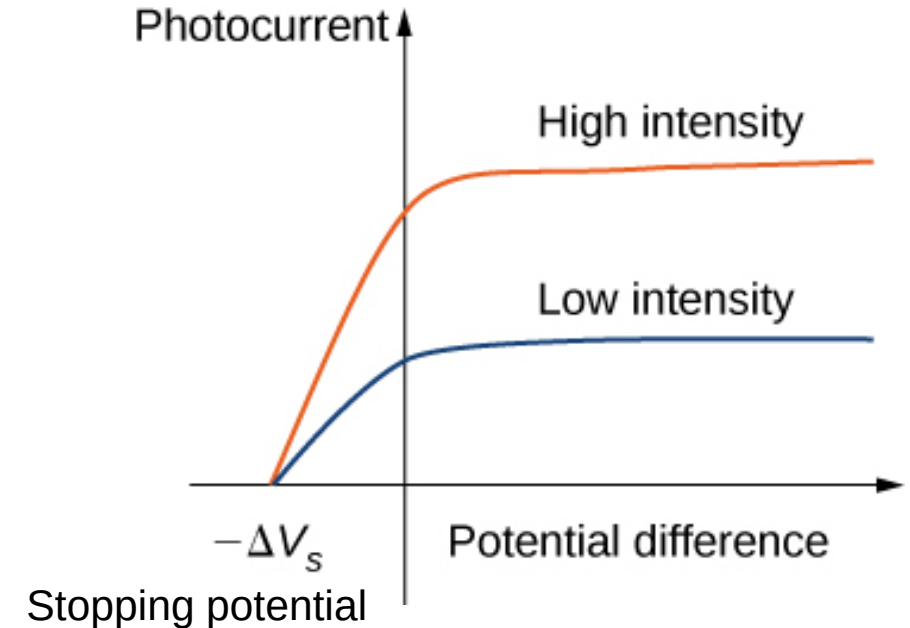


The photo-electric effect

Observations

Effect of intensity

- If $\nu < \nu_c$, no electron emission (hence no photocurrent) can be observed.
- If $\nu > \nu_c$, electron emission takes place and photocurrent increases with light intensity
 - More intensity means more photons having energy sufficient enough to produce photoelectron.
 - More photoelectron means more photocurrent.



The stopping potential (or retarding potential) in the photoelectric effect remains constant regardless of light intensity because the kinetic energy of the emitted photoelectrons depends solely on the frequency (or wavelength) of the incident light and not on its intensity. While intensity affects the number of photoelectrons emitted (i.e., the photoelectric current), it doesn't alter the energy of the individual photoelectrons.

The photo-electric effect

Observations

Effect of Frequency

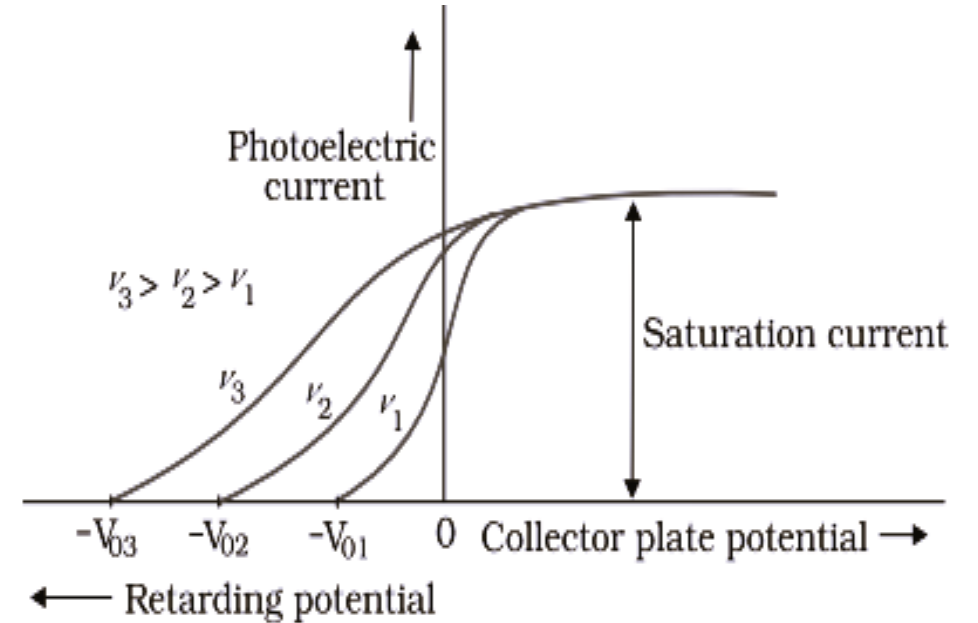
Frequency determines the energy of individual photons:

$$E = hf$$

$hf < \phi$ (work function), no electrons are emitted $\Rightarrow$ no photocurrent.

If $hf > \phi$, electrons are emitted with kinetic energy:

$$KE = hf - \phi$$



The stopping potential (or retarding potential) increases with increasing frequency.

At some point, all the emitted electrons (regardless of their initial direction or energy) are already being collected. Further increasing the potential doesn't change the number of emitted electrons — it only attracts them faster. Since photocurrent depends on the number of electrons collected per unit time, once all are being collected, the current stabilizes.

The photo-electric effect

Observations

Effect of metal

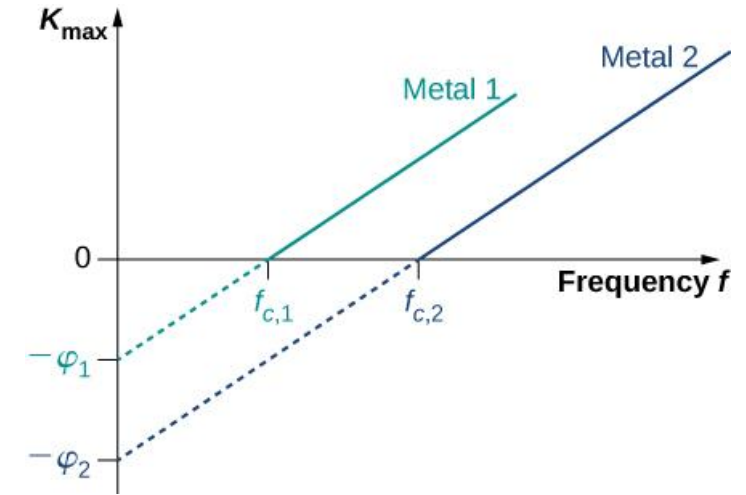
If $hf > \phi$, electrons are emitted with kinetic energy:

$$KE = hf - \phi$$

The equation is similar to

$$y = mx + c$$

Different metals have different work function and hence the curve



1. Ultraviolet radiation has a frequency of 6.8×10^{15} Hz. Calculate the energy, in joules, of the photon.
2. A sodium vapor lamp emits light photons with a wavelength of 5.89×10^{-7} m. What is the energy of these photons?
3. One of the electron transitions in a hydrogen atom produces infrared light with a wavelength of 746.4 nm. What amount of energy causes this transition?
4. What is the frequency of UV light that has an energy of 2.39×10^{-18} J?